

DES-006 cobe: expert-review iteration

DRAFT / PROTOTYPE, 2026-09-25. cobe: classical barrel endcap (formerly A). Only cobe and pint remain active. Cylindrical short strips; ODD-guided control.

Geometry and motivation

Retain the tracker host and $|\eta| < 4$ requirement. Separate active pixel, short-strip and long-strip disk bands with provisional 10 mm gaps. Move the outer long-strip barrels and disks together to recover leverage around $\eta=1.1$. No component or geometry is signed off.



Pixel barrel radii: 34, 60, 106, 182 mm; half-length 550 mm. Short-strip central radii: 260, 340, 480, 660 mm. Long-strip radii: 840, 1060 mm; half-length 1400 mm. First long-strip disk: $z=1430$ mm. Exact signed layers are in DES-006-named-layouts.json.

Disk annuli: pixels 35-190 mm; short strips 200-700 mm; long strips 710-1100 mm. Those active bounds reserve space but do not prove that real supports or services fit. A 25 mm pipe-exclusion plus 5 mm installation fixture is conditional; the 34 mm first radius is not a cleared interface.

cobe: strengths, weaknesses and next decision

Demonstrated prototype benefit

At $\eta=1.1$, origin, 3 T, the Gaussian q/p_T uncertainty improves 5.7% at $p_T=1$ GeV and 8.5% at 100 GeV relative to original A (cobe control). The generic five-parameter covariance uses straight-reference crossings and leading thin-scatterer noise. It is not a reconstructed resolution or a full material budget.

Ideal silicon areas: pixel 4.612 m²; short strip 43.203 m²; long strip 120.077 m² including both faces. Long-strip pair material is counted once. Earlier controls remain in the original files.

Costs and unresolved weaknesses

The worst finer-grid q/p_T uncertainty ratio is 1.430, so the landmark improvement is not uniform. Narrower disks lose hits in some transitions. The 60 mm second pixel layer trades better high- p_T IP information against about 4% worse central low- p_T d_0 . Far-forward leverage remains limited. These losses prevent adopting the full iterated layout as an optimum.

cobe remains the simplest geometric control. The longer outer barrel increases the area and changes the service handoff. IdRes was run on cobe only; its zero-material limit agrees with the independent covariance at printed precision.

Priorities and human review

Review beam-pipe and service envelopes first; repair lost transition coverage with real module masks; agree beamspot/performance weights; revisit pint row count and tiling; then run ACTS and later full simulation. Source locators, all attempted scans, adverse cases and individual comment responses are in DES-006-expert-review.md. Final sign-off remains with asalzburger-review.